Wavelet-Enhanced Spatial–Spectral Prior Injection Mamba Network for Hyperspectral Image Super-Resolution

Guohua Miao, *Graduate Student Member, IEEE*, Zhihua Xie, *Member, IEEE*, and Fan Yang

***Abstract*—Deep learning-based methods, particularly Mamba-based models, hold huge potential for hyperspectral image super-resolution (HSI-SR) reconstruction. However, the intrinsic spatial–spectral details in HSIs are hard to be completely represented on insufficient data solely utilizing the hierarchical learning manner in most Mamba structures, which results in the degradation of restoration quality on inherent fine-grain textures. To address this challenge, this article proposes a wavelet-enhanced spatial–spectral prior guided mamba network (WESSP-Mamba) for HSI-SR, which leverages wavelet-enhanced spatial–spectral priors (SSPriors) to refine detail recovery with cross-layer embedment. Specifically, this model strengthens multiscale spatial correlations with a Wavelet transform and produces spectral responses with a spectral attention mechanism, which adaptively highlights specific bands to acquire an instructive spatial–spectral joint prior. The learnable prior is then injected into the hierarchical deep features from the backbone network—built upon a shuffle–reshuffle mamba (SRM) module—through a spatial selection mechanism, thereby contributing to the impressive representation capability. Extensive experiments on three representative datasets show that the WESSP-Mamba model achieves superior HSI-SR performance to other deep learning approaches in terms of quantitative and qualitative metrics. Especially, the peak signal-to-noise ratio (PSNR) of WESSP-Mamba exceeds that of the second-best method by 0.0608, 0.0520, 0.0443 dB on the Chikusei, Houston, and Pavia Centre datasets at the $\times 4$ scale factor, respectively.**

***Index Terms*—Hyperspectral image super-resolution (HSI-SR), mamba, spatial selection mechanism, spatial–spectral priors (SSPriors), wavelet transform.**

I. Introduction

HYPERSPECTRAL image (HSI), involving continuous and abundant spectral imagery with a wide coverage range, can reflect subtle response differences and material characteristics of targets across spectral bands [1], which has potential application value in fields such as object classification [2], [3], environmental monitoring [4], [5], and medical imaging [6], [7]. However, due to hardware limitations, the high spectral resolution often compromises spatial resolution [8], which in turn undermines the wide applications of HSI in downstream tasks. To alleviate the practical bottleneck caused by the relatively low spatial resolution, HSI super-resolution (HSI-SR) reconstruction technology has received remarkable attention in recent decades. This methodology aims to reconstruct high-resolution (HR) images with both high spatial definition and spectral fidelity from low-resolution (LR) HSIs, thereby extraordinarily extending the potential value of hyperspectral imagery.

Recently, efforts based on deep networks have emerged as a promising research hotspot for HSI-SR. Generally, mainstream works employ convolutional neural networks (CNNs) or Transformer-based architectures as the backbone, such as SSPSR [9], MSFMNet [10], MSDformer [11], and CTUNet [12]. Although these solutions have shown encouraging performance, the intrinsic high dimensionality and redundancy of hyperspectral data still pose serious obstacles for the precise SR reconstruction. It is well known that CNNs cannot effectively model long-range spatial and spectral dependencies. In contrast, Transformer offers strong global modeling capabilities through self-attention mechanisms. However, the quadratic computational complexity restricts their scalability to high-dimensional HSI data.

Fortunately, Mamba structure, built on state space models, has attracted considerable attention due to its linear computational complexity and powerful global modeling capabilities [13]. Mamba was originally developed for natural language processing, and has recently been extended to 2-D image modeling by Zhu et al. [14] and Liu et al. [15], exhibiting promising merits. To abate the computational burden of processing large-scale data, researchers have attempted to explore the application of Mamba in SR tasks [15], [16], [17]. Traditionally, these methods transform an HSI into a 1-D sequence for modeling via bidirectional or four-directional scanning. Nevertheless, this flattening process compromises the inherent spatial structures in HSI. More importantly, due to the causal nature of Mamba, these works struggle to capture comprehensive spatial dependencies [18].

Obviously, existing HSI-SR methods based on Mamba or Transformer usually exploit a cascaded feature extraction and reconstruction framework [11], [12], [16], [19], [20], where spatial and spectral details inevitably degrade progressively

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Data is available on-line at https://github.com/mDaHua/WESSP-Mamba.git
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as the network deepens. As shallow structural information gradually weakens, some high-frequency textures and edge details will be hard to maintain. This loss of these subtle features not only compromises the accurate restoration of spatial structures but also results in spectral distortion. Especially for Mamba networks, which require flattening images into 1-D sequences and performing causal modeling, the original structural information is vulnerable to degradation. Based on this observation, Mamba's modeling capabilities in HSI-SR remain insufficient to ensure the completeness of fine details and the structural consistency of spatial–spectral dependence.

To overcome the aforementioned issues, this article proposes a novel HSI-SR model based on the Mamba network, named the wavelet-enhanced spatial–spectral prior guided Mamba super-resolution network (WESSP-Mamba). The core idea of this approach lies in constructing a cross-layer interactive prior injection mechanism, where the spatial–spectral prior (SSPrior) acts as a learnable structural guidance, deeply coupling with the hierarchical modeling process of Mamba to compensate for the detail loss during the Mamba-based hierarchical learning. Specifically, the model mainly comprises a backbone framework and a prior branch. The backbone employs a cascaded structure to progressively model deep features, utilizing Mamba modules and a spectral attention mechanism to capture spatial–spectral dependencies. To alleviate the limitations of causal modeling inherent in Mamba, we introduce a shuffle-based structural rearrangement strategy, enabling the model to capture diverse global spatial dependencies. For prior information acquisition, we exploit the powerful frequency domain separation capability of the discrete wavelet transform (DWT) [21], [22] to guide the prior modeling. By incorporating spatial and spectral attentions, the module adaptively adjusts information across different frequency bands, thereby capturing more discriminative frequency components and refining details that are easily overlooked in the hierarchical learning paradigm. Unlike previous approaches [9], [17], we directly generate learnable SSPrior information from the original LR-HSI via the combination of DWT and learnable CNN, and adopt a "single-extraction, multi-injection" strategy that progressively integrates the extracted prior into the backbone's deep features through a spatial selection mechanism. This strategy not only mitigates potential information loss during hierarchical modeling but also avoids the additional computational overhead associated with excessive DWT operators for feature representation, thereby substantially boosting the reconstruction fidelity of HSIs. In summary, the main contributions of this article are as follows.

1) Built upon the Mamba framework as the backbone, we incorporate spatial and spectral prior information to interactively compensate for the insufficient learning ability on limited complex HSI data, facilitating the diverse details restoration.
2) We design a prior extraction module that combines DWT and spectral attention to comprise spatial and spectral priors, supplementing the intricate textures that can be discarded in a purely progressive learning paradigm.
3) We construct a shuffle–reshuffle mamba (SRM) module that integrates a shuffle-based feature enhancement mechanism to aggregate the modeling of global spatial dependencies.

The remainder of this article is organized as follows. Section II provides a brief review of existing super-resolution methods; Section III presents a detailed description of our proposed method; Section IV reports experimental design, comparative results, and ablation studies; Section V concludes the article.

II. Related Work

A. CNN-Based Super-Resolution Methods

In the last decade, CNN-based methods have been extensively explored for HSI-SR owing to their strong feature extraction capability. Zheng et al. [23] implemented cascaded dense residual blocks and combined an L1 loss and a spectral loss to enhance spatial resolution, while preserving spectral consistency. Li et al. [24] built grouped recursive models to decrease the complexity and improve reconstruction accuracy. Jiang et al. [9] employed grouped convolution with progressive upsampling and attention mechanisms to emphasize spatial–spectral feature extraction. However, the grouped convolution operator may incur partial spectral information loss. To capture the texture and structural features of HSI, Yang et al. [25] designed a 3-D convolution-based wavelet packet prediction network for reconstruction. Wang et al. [26] explored wavelet decomposition with a three-branch D CNN to acquire frequency features and improve reconstruction quality. Nonetheless, such approaches are still subject to the inconsistency of the Heisenberg uncertainty principle. Subsequently, Prasad et al. [27] exploited D separable convolutions for the SR task. Yang et al. [28] integrated local and nonlocal D attention to capture contextual information and long-range dependencies in HSI. Li et al. [29] investigated the potential of Kolmogorov–Arnold networks (KANs) in modeling long-range dependencies across the spatial and spectral dimensions of HSI. These studies reveal the effectiveness of CNNs in spatial–spectral feature modeling. Nevertheless, the insufficient receptive field and difficulty in capturing long-range dependencies constrain their ability to fully mine global contextual relations, which motivates the exploration of Transformer-based schemes.

B. Transformer-Based Super-Resolution Methods

To tackle the incapability of CNNs to learn long-range dependencies, Transformer-based frameworks have been incorporated into HSI-SR research. Chen et al. [11] integrated dilated convolutions with a deformable convolution-based Transformer to extract multiscale spatial–spectral features and capture global dependencies. Chen et al. [30] achieve long-range dependencies with cross-attention across spatial and spectral dimensions, exploring synergy between global features. Wu et al. [31] combined low-complexity global self-attention with window self-attention, achieving efficient feature modeling with reduced complexity. Zhang et al. [32] developed a sparse spectral transformer module that explores global correlations and long-range dependencies across spectral bands through a coarse-to-fine strategy. Although these

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studies have demonstrated the excellent global modeling capability of Transformer. However, these methods still face challenges in inadequately balancing the contributions of local details and global contextual features, and often exhibit unsatisfactory hierarchical information interaction. To address this issue, Zhang et al. [33] combined a dense cross-scope transformer with a weighted connection and incorporated a wavelet transform to reduce information loss and strengthen fine details. Similarly, to dig out high-frequency details, Zhang et al. [34] aggregated spatial and channel self-attention to capture global features and utilized a frequency-gated feedforward network to amplify high-frequency signals and suppress low-frequency noise, which significantly refined the detailed representation. Nevertheless, the efficiency of cross-domain feature fusion and collaborative modeling requires further optimization. To provide higher quality network inputs, Zhang et al. [19] obtained auxiliary prior features with a diffusion model, followed by fusion and aggregation modules for fine deep spatial–spectral information. Despite Transformer variants having achieved notable improvement in HSI-SR, their quadratic computational complexity and heavy computational overload necessitate relatively lightweight alternatives.

C. Mamba-Based Super-Resolution Methods

With the emergence of Mamba, researchers have probed its potential in HSI-SR. To reduce computational burden, Zhi et al. [16] trained a lightweight network in combination with Mamba for long-range spatial modeling and Transformer for spectral feature extraction, achieving strong reconstruction with fewer parameters. Lin et al. [35] employed bidirectional cross-scanning and cross-attention mechanisms to produce spatial–spectral representation, mitigating the information loss caused by decomposing 3-D data into 1-D sequences. Zhu et al. [15] effectively improve image detail reconstruction by combining local features extracted by CNN with global features captured by Mamba. Xu et al. [36] introduced spatial and spectral Mamba modules with four- and bidirectional scanning, respectively, to model long-range dependencies in HSI. Li et al. [20] designed a three-stage Mamba-based architecture that combines multiscale convolution, channel shuffle, and attention mechanisms to progressively extract spatial features, enhance spatial–spectral fusion, and refine spectral details at a low computational burden. Xiao et al. [17] injected Fourier-domain features into Mamba and its gated attention-linear module to improve global modeling and feature refinement. Chen et al. [37] integrated a local spatial–spectral partitioning and a global spectral reordering into Mamba, mitigating the local forgetting issue and inherent sensitivity to input sequences. Taken together, Mamba-based paradigms exhibit massive advantages in feature modeling capability and computational efficiency. Ensembling Mamba with other modules in various ways, it enables flexible network designs and provides a promising pipeline for HSI-SR.

III. Methodology

In this section, we present a detailed description of the proposed WESSP-Mamba framework. The discussion is organized into the following subsections: the overall network architecture, the prior-guided spectral shuffle-mamba block (PGSSMB), the SSPrior block, and the prior fusion block (PFB). Additionally, we elaborate on the loss functions utilized in training the WESSP-Mamba model.

A. Overall Network Architecture

As mentioned above, most HSI-SR models employ a cascaded backbone architecture to progressively model spatial–spectral relations from local to global levels. However, such a layer-by-layer feature transmission paradigm inevitably leads to the gradual degradation of crucial but subtle details. This issue is further exacerbated in Mamba-based methods, where the transformation of HSI into 1-D sequences tends to suffer from the loss of spatial–spectral causal features. To cope with this problem, we incorporate SSPrior information—learning directly from the original HSI—into the cascaded phases of the main network, which helps the model refocus on critical details that may be overlooked. Given the strong capability of Wavelet transforms in multiscale analysis and detail modeling, this work employs a DWT-based method in combination with a spectral attention mechanism to explore SSPriors. Briefly, our model aims to aggregate the network's ability to model fine-grained details by injecting learnable informative priors based on DWT. The overall framework of the WESSP-Mamba model is illustrated in Fig. 1, which mainly includes four components: the shallow feature extraction, the SSPrior extraction, the deep feature extraction, and the image reconstruction.

For convenience of description, we define the input as $X \in \mathbb{R}^{\mathrm{B}\times h\times w}$ and the output as $Y \in \mathbb{R}^{\mathrm{B}\times H\times W}$, where B is the number of spectral bands, $h \times w$ denotes the spatial resolution of the LR image, and $H \times W$ corresponds to the HR output, where $H = s \times h$ and $W = s \times w$, s represents the SR scale factor. The SR reconstruction process of WESSP-Mamba can be formulated as

$$Y = H_{\text{WESSP-Mamba}}(X) \tag{1}$$

where $H_{\text{WESSP-Mamba}}(\cdot)$ denotes the mapping function of the proposed model.

In the shallow feature extraction stage, we first apply a convolutional operation to the input LR-HSI X, projecting it into a feature space with C channels to enhance feature representation. Then, a residual convolutional module is used to obtain shallow feature representations. This process can be expressed as

$$F_0' = \text{Conv}(X) \tag{2}$$

$$F_0 = \sigma\left(\text{Conv}\left(\text{Conv}\left(F_0'\right)\right)\right) + F_0' \tag{3}$$

where $F_0' \in \mathbb{R}^{C\times h\times w}$ denotes the intermediate features after the initial convolution, and $F_0 \in \mathbb{R}^{C\times h\times w}$ represents the final output of the shallow feature extraction.

In the SSPrior extraction stage, we adaptively acquire feature maps from the input HSI X through multiscale decomposition based on DWT and CNN-based texture refinement, which retains as much of the meaningful information as possible. These extracted prior features are then projected

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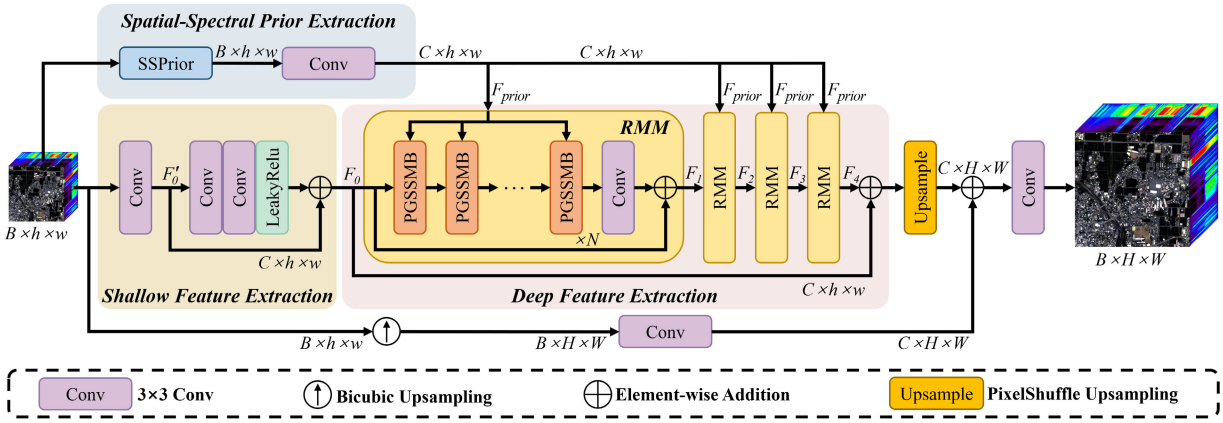


Fig. 1. Overall framework of the proposed network.

into C-dimensional space before being fed into subsequent modules. This process can be formulated as

$$F_{\text{prior}} = \text{Conv}\left(H_{\text{Prior}}\left(\mathbf{X}\right)\right) \tag{4}$$

where $F_{\text{prior}} \in \mathbb{R}^{C\times h\times w}$ represents the final features obtained from the prior branch, and $H_{\text{Prior}}(\cdot)$ denotes the SSPrior extraction operation.

In the deep feature extraction stage, four cascaded residual mamba modules (RMMs) are used to progressively mine features, with global residual connections to preserve structural information. Each RMM has two inputs, $F_i, i \in \{0,1,2,3\}$ and F_{prior}. This process can be expressed as

$$F_{i+1} = H_{\text{RMM}_{i+1}}\left(\ldots H_{\text{RMM}_i}\left(F_0, F_{\text{prior}}\right), \ldots, F_{\text{prior}}\right) + F_0 \tag{5}$$

where $F_{i+1} \in \mathbb{R}^{C\times h\times w}$ denotes the $(i+1)$th output of the RMM module. Each RMM is composed of N PGSSMBs followed by a convolutional layer, then incorporates module-level residual connections to prevent the loss of subtle information. The process of $H_{\text{RMM}_{i+1}}(\cdot)$ can be formulated as

$$F_i' = H_{\text{PGSSMB}_N}\left(\ldots H_{\text{PGSSMB}_1}\left(F_i, F_{\text{prior}}\right), \ldots, F_{\text{prior}}\right) \tag{6}$$

$$F_{i+1} = \text{Conv}\left(F_i'\right) + F_i \tag{7}$$

where $H_{\text{PGSSMB}_N}(\cdot)$ denotes the Nth PGSSMB block within the RMM.

In the image reconstruction phase, the deep features are upsampled, and interact with the original input image via a global residual connection. Finally, a convolutional operation is applied to acquire the HR output image Y. This reconstruction process can be represented as

$$Y = \text{Conv}\left(\text{up}\left(F_4\right)^p + \text{Conv}\left(\text{up}\left(X\right)^b\right)\right) \tag{8}$$

where $\text{up}(\cdot)^p$ denotes pixel shuffle upsampling and $\text{up}(\cdot)^b$ denotes bicubic upsampling of the input image.

B. Prior-Guided Spectral Shuffle-Mamba Block

To comprehensively fuse spatial–spectral features while effectively leveraging prior information, we design the PGSSMB module, as shown in Fig. 2. Aiming to boost Mamba's capacity for capturing spatial–spectral dependencies, this work incorporates a targeted enhancement strategy into the visual state space (VSS) blocks. Although the four-directional scanning strategy partially alleviates the limitations of its receptive field and enhances contextual modeling, its ability to capture global spatial dependencies and feature interactions remains insufficient. Inspired by the endeavor in [38], we inject a spatial shuffle branch into the original Mamba in a cross-layer manner. This branch rearranges spatial features to facilitate interfeature learning, thus promoting the model's representation of spatial structures.

To further aggregate spectral modeling, a spectral attention mechanism (SpeA) is integrated in parallel within the core of PGSSMB. This mechanism highlights informative spectral bands, strengthening the network's capacity to detect interband dependencies. Additionally, to promote the module's nonlinear representational power, a multilayer perceptron (MLP) is employed perform feature transformation and produce more expressive deep features. The resulting deep features and the SSPrior are jointly fed into the PFB module to collaboratively obtain detailed restoration. The detailed mechanism of PFB will be elaborated in Section III-D. Moreover, to increase the network's adaptability, we introduce a residual modulation coefficient that dynamically adjusts the intensity of residual fusion.

Given an input feature F_i to PGSSMB, we first apply layer normalization. The normalized feature is then processed simultaneously by the SpeA and SRM modules to extract spectral and spatial representations, respectively. The resulting features are fused and then integrated with the input via a residual connection to preserve intricate information. This process can be formulated as

$$F' = \text{LN}\left(F_i\right) \tag{9}$$

$$F'' = H_{\text{SpeA}}\left(F'\right) + H_{\text{SRM}}\left(F'\right) + F_i \odot \alpha 1 \tag{10}$$

where $\text{LN}(\cdot)$, $H_{\text{SpeA}}(\cdot)$, and $H_{\text{SRM}}(\cdot)$ denote the processing procedures of layer normalization, the SpeA, and the SRM, respectively, and $\alpha 1$ is the residual scaling coefficient. Subsequently, the resulting features are sequentially processed by an MLP and the PFB. This overall process can be represented

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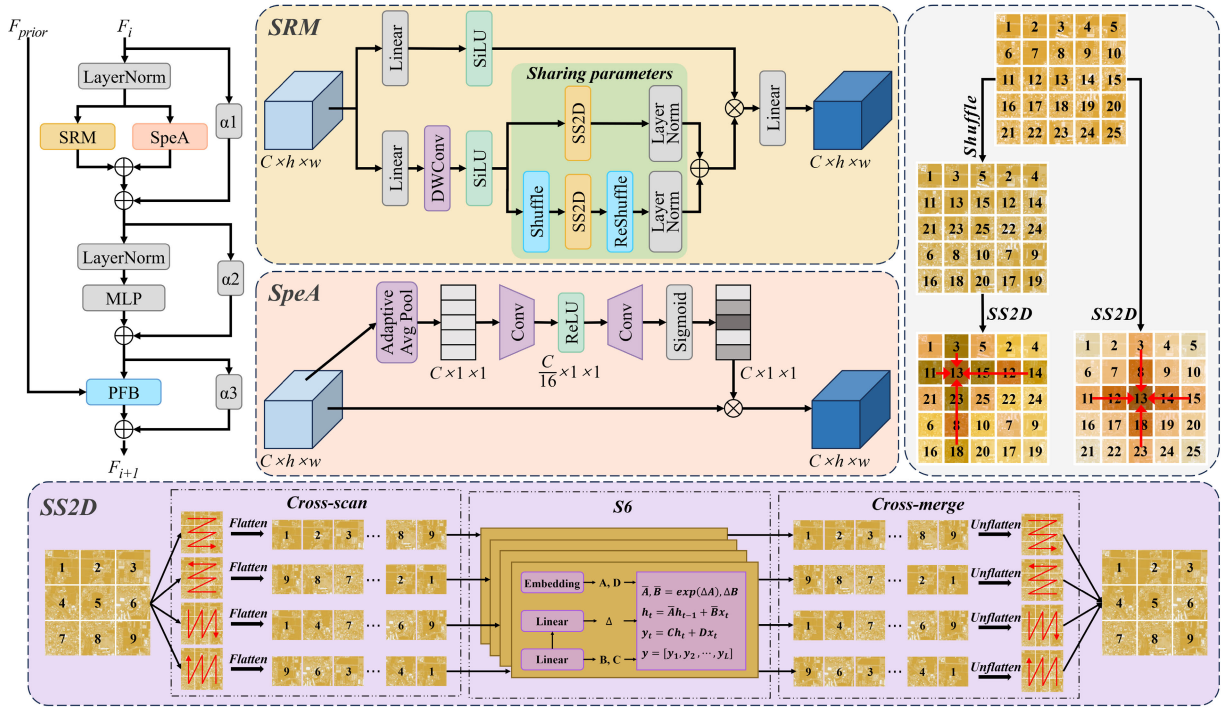


Fig. 2. Diagram of PGSSMB.

by the following equation:

$$F''' = \mathrm{MLP}\left(\mathrm{LN}\left(F''\right)\right) + F'' \odot \alpha 2 \tag{11}$$

$$F_{i+1} = H_{\mathrm{PFB}}\left(F''', F_{\mathrm{prior}}\right) + F''' \odot \alpha 3 \tag{12}$$

where $\mathrm{MLP}(\cdot)$ and $H_{\mathrm{PFB}}(\cdot,\cdot)$ denote the processing procedures of the MLP and PFB, respectively. $\alpha 2$ and $\alpha 3$ are residual scaling coefficients.

In the SpeA module, the input feature F' first undergoes an adaptive average pooling along the spatial dimensions to yield a global response from each channel, resulting in a channel-level descriptor vector. This vector is then passed through a bottleneck structure consisting of two fully connected layers for nonlinear transformation. Finally, a Sigmoid function generates spectral attention weights, which are multiplied channel-wise with the original feature to concentrate on informative spectral bands. The process is described as

$$F_{\mathrm{spe}} = \mathrm{Sig}\left(\mathrm{Conv}\left(\delta\left(\mathrm{Conv}\left(\mathrm{Pool}\left(F'\right)\right)\right)\right)\right) \odot F' \tag{13}$$

where $F_{\mathrm{spe}} \in \mathbb{R}^{C\times h\times w}$ is the output of $H_{\mathrm{SpeA}}(\cdot)$, $\delta(\cdot)$ denotes the ReLU activation function, $\mathrm{Sig}(\cdot)$ is Sigmoid function, and $\mathrm{Pool}(\cdot)$ is the adaptive average pooling operation.

The SRM module broadly comprises two parallel components: a gating-signal generation branch and a feature-extraction branch. Different from the vanilla VSS block, the SRM module builds an additional parallel feature enhancement branch based on a global cross-disruption strategy within the feature extraction pathway. This design enables the model to not only capture local relationships between adjacent pixels but also to account for long-range dependencies across distant pixels, whose details are visualized in Fig. 2. Under the fixed four-directional scanning scheme, correlations between the first and last elements can only be established after traversing all pixels. In contrast, the perturbation strategy allows the model to more efficiently capture potential dependencies among pixels at diverse spatial positions. Specifically, in the gating-signal generation branch, the features are first projected to a space of αC dimensions and activated to generate gating weights, where α denotes the block expansion factor. This process can be expressed as

$$z = \varphi\left(\mathrm{Linear}\left(F'\right)\right) \tag{14}$$

where $\mathrm{Linear}(\cdot)$ is a linear mapping operation, $\varphi(\cdot)$ is the sigmoid-weighted linear unit (SiLU) activation function, and z is the generated gating signal. In the feature-extraction branch, the feature is similarly projected to αC dimensions, then processed through a convolution filter, SiLU activation function, and two weight-sharing submodules to obtain global features. Among these submodules, one applies standard SS2D, while the other follows a Shuffle–SS2D–ReShuffle sequence. The outputs of the two submodules are normalized and fused by element-wise addition to yield the output of the feature-extraction branch

$$F'_{\mathrm{S2D}} = \varphi\left(\mathrm{DWConv}\left(\mathrm{Linear}\left(F'\right)\right)\right) \tag{15}$$

$$F_{\mathrm{S2D}} = H_{\mathrm{S2D}}\left(F'_{\mathrm{S2D}}\right) + H_{\mathrm{RS}}\left(H_{\mathrm{S2D}}\left(H_{S}\left(F'_{\mathrm{S2D}}\right)\right)\right) \tag{16}$$

where $\mathrm{DWConv}(\cdot)$ is the depthwise separable convolution, F_{S2D} represents the feature-extraction branch's output feature. $H_{\mathrm{S2D}}(\cdot)$, $H_{S}(\cdot)$ and $H_{\mathrm{RS}}(\cdot)$ denote the SS2D, Shuffle, and ReShuffle operations, respectively. Finally, F_{S2D} is modulated by the gating signal z and projected back to the original C

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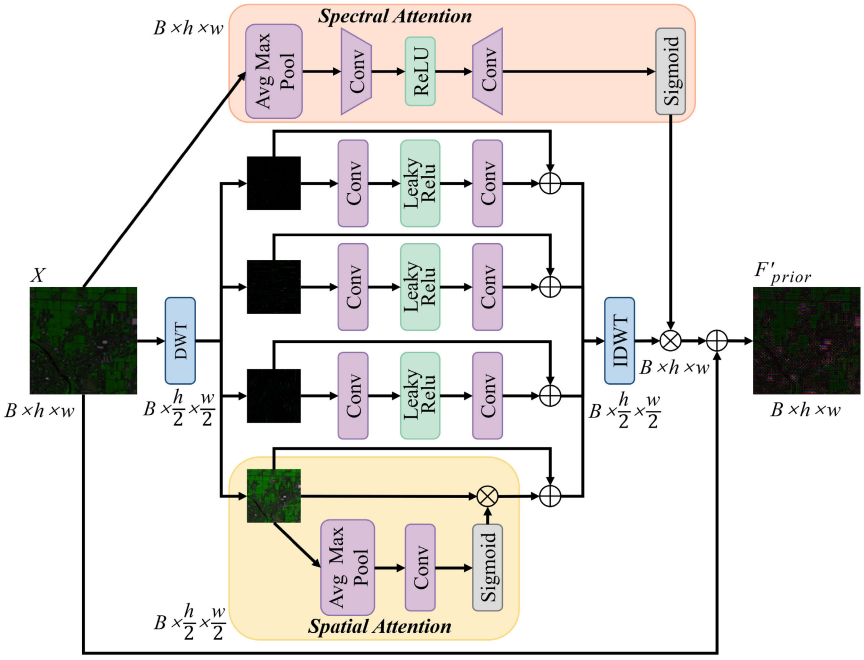


Fig. 3. Diagram of SSPrior block.

dimension to obtain the final output of the SRM module. This process can be represented as

$$F_{\mathrm{SRM}} = \mathrm{Linear}\left(F_{\mathrm{S2D}} \otimes z\right) \tag{17}$$

where $F_{\mathrm{SRM}} \in \mathbb{R}^{C\times h\times w}$ is the final output of $H_{\mathrm{SRM}}(\cdot)$.

C. SSPrior Block

The detailed structure of the SSPrior block is depicted in Fig. 3. To make full use of the spatial–spectral nature of DWT [39], the core motivation of this block is that suitable treatment strategies are undergone on specific frequency components in prior information generation. For the low-frequency components, we incorporate a spatial attention mechanism for representation on expressive structural regions. In contrast, for the high-frequency components, convolutional operations are applied to concentrate on diverse textures and edge details, thereby accomplishing more targeted feature modeling. Additionally, learnable spectral attention weights highlight spatial features across channels, facilitating joint spatial–spectral representation. To further promote the stability and fidelity of prior representation, a global residual connection is embedded into the entire module to mitigate potential information loss during the modeling stage.

In the spatial feature extraction part, the HSI X is transformed into four frequency subbands via DWT, which can be expressed as

$$\mathrm{LL}, \{\mathrm{LH}, \mathrm{HL}, \mathrm{HH}\} = \mathrm{DWT}\left(X\right) \tag{18}$$

where $\mathrm{LL}, \{\mathrm{LH}, \mathrm{HL}, \mathrm{HH}\} \in \mathbb{R}^{\mathrm{B}\times h/2\times w/2}$ represents the low-frequency component, and the high-frequency components in the vertical, horizontal, and diagonal directions, respectively. $\mathrm{DWT}(\cdot)$ denotes the DWT operation.

For the low-frequency component, we introduce a residual spatial attention mechanism to automatically adjust the attention to spatial regions. This process is expressed as

$$F'_{\mathrm{LL}} = \mathrm{Sig}\left(\mathrm{Conv}\left(\mathrm{Cat}\left(\mathrm{mean}\left(\mathrm{LL}\right), \max\left(\mathrm{LL}\right)\right)\right)\right) \tag{19}$$

$$F_{\mathrm{LL}} = F'_{\mathrm{LL}} \odot \mathrm{LL} + \mathrm{LL} \tag{20}$$

where $F_{\mathrm{LL}} \in \mathbb{R}^{\mathrm{B}\times h/2\times w/2}$ is the processed feature of the low-frequency component, $\mathrm{Cat}(\cdot,\cdot)$ denotes channel concatenation. And $\mathrm{mean}(\cdot)$ and $\max(\cdot)$ refer to average pooling and max pooling, respectively. For the high-frequency components, we use residual convolution operations for all three bands, which can be described as

$$F_{hf} = \mathrm{Conv}\left(\sigma\left(\mathrm{Conv}\left(hf\right)\right)\right) + hf, \quad hf \in \{\mathrm{LH}, \mathrm{HL}, \mathrm{HH}\} \tag{21}$$

where $F_{hf} \in \mathbb{R}^{\mathrm{B}\times h/2\times w/2}$ denotes the processed high-frequency features. Subsequently, the four frequency-domain features are reconstructed into a spatial-domain representation using the inverse DWT (IDWT)

$$F_{\mathrm{pspa}} = \mathrm{IDWT}\left(F_{\mathrm{LL}}, \{F_{\mathrm{LH}}, F_{\mathrm{HL}}, F_{\mathrm{HH}}\}\right) \tag{22}$$

where $F_{\mathrm{pspa}} \in \mathbb{R}^{\mathrm{B}\times h/2\times w/2}$ denotes the reconstructed spatial features after wavelet processing, and $\mathrm{IDWT}(\cdot,\cdot)$ represents the inverse wavelet transform operation.

In the spectral information extraction part, the HSI X is processed by a SpeA to obtain attention weights along the spectral dimension, which are then fused with F_{pspa} through interaction. A global residual connection is then applied to obtain the final SSPrior representation. This process is formulated as

$$F'_{\mathrm{prior}} = \mathrm{Sig}\left(\mathrm{Conv}\left(\delta\left(\mathrm{Conv}\left(\mathrm{Pool}\left(X\right)\right)\right)\right)\right) \odot F_{\mathrm{pspa}} + X \tag{23}$$

where $F'_{\mathrm{prior}} \in \mathbb{R}^{\mathrm{B}\times h\times w}$ is the output of the $H_{\mathrm{Prior}}(\cdot)$ function.

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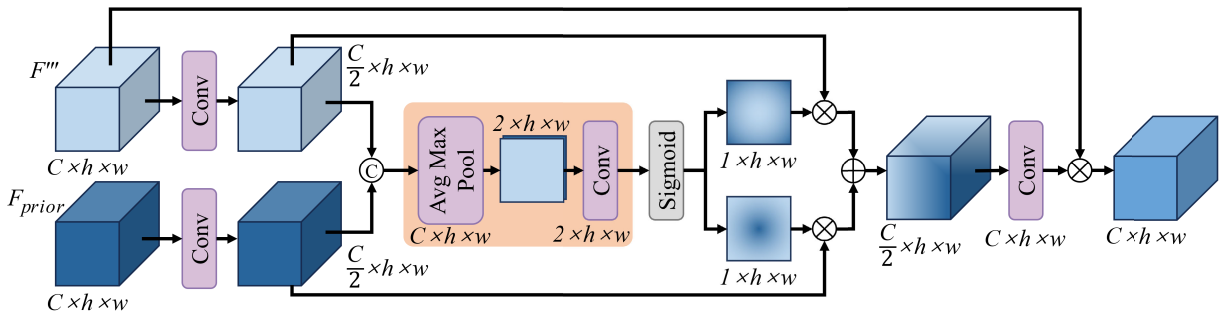


Fig. 4. Diagram of PFB.

D. Prior Fusion Block

To adaptively select and integrate diverse information from both prior features and deep features, we design a PFB inspired by the spatial selection mechanism [40], as illustrated in Fig. 4.

The PFB module receives the deep feature F''' and the prior feature F_{prior} as inputs. Since prior and deep features may contain considerable redundant information, direct concatenation or addition can lead to noise accumulation. To mitigate this, the PFB first compresses the input features to extract high-level representations

$$F_{m1} = \text{Conv}\left(F'''\right), \quad F_{m2} = \text{Conv}\left(F_{\text{prior}}\right) \tag{24}$$

where $F_{m1} \in \mathbb{R}^{C/2\times h\times w}$ and $F_{m2} \in \mathbb{R}^{C/2\times h\times w}$ are the high-level features representations after convolution. To enable effective complementarity between the two feature streams, a spatial attention mechanism is introduced to adaptively modulate the high-level features. Specifically, the compressed features are concatenated and subjected to both global average pooling and max pooling to capture salient information from different statistical perspectives. The resulting descriptors are then processed by lightweight convolution layers and nonlinear mappings to generate two spatial attention maps, which are applied to the prior and deep features, respectively, to enhance or suppress their responses. Subsequently, they are fused via element-wise addition. This process is defined as

$$F_c = \text{Cat}\left(F_{m1}, F_{m2}\right) \tag{25}$$

$$w_1, w_2 = \text{sp}\left(\text{Sig}\left(\text{Conv}\left(\text{Cat}\left(\text{mean}\left(F_c\right), \max\left(F_c\right)\right)\right)\right)\right) \tag{26}$$

$$F_w = F_{m1} \odot w_2 + F_{m2} \odot w_2 \tag{27}$$

where $F_c \in \mathbb{R}^{C\times h\times w}$ represents the initial fused features, $w_1 \in \mathbb{R}^{1\times h\times w}$ and $w_2 \in \mathbb{R}^{1\times h\times w}$ are the two attention weights, $F_w \in \mathbb{R}^{C/2\times h\times w}$ represents the fused features, and sp$(\cdot)$ denotes the channel-wise split operation. Finally, the fused features are aligned through a convolution layer and combined with the original deep features, preserving the semantic richness of the deep representations while enhancing spatial details, thus offering more robust and informative features for the subsequent reconstruction component

$$F_{\text{PFB}} = \text{Conv}\left(F_w\right) \odot F''' \tag{28}$$

where $F_{\text{PFB}} \in \mathbb{R}^{C\times h\times w}$ denotes the output of the PFB.

E. Loss Function

Based on previous studies [12], [30], a composite loss function, including the l_1 loss, spectral angle mapper (SAM) loss, and gradient loss, was designed to thoroughly assess the performance of SR reconstruction. Following this practice, this work also adopts a similar composite loss function to simultaneously measure the spectral consistency, spatial structure clarity, and overall perceptual quality of the restored HSI.

Specifically, the l_1 loss, which penalizes pixel-wise differences between the reconstructed and reference images, is effective in preserving edges and structural details while reducing oversmoothing. The SAM loss measures the angular deviation between spectral vectors, thereby enhancing spectral consistency between the reconstruction and the ground truth. Moreover, the gradient loss captures spatial variation trends, improving the recovery of edges and textures, and thus emphasizing the perceptual quality.

Accordingly, the total loss function of our model is a weighted fusion from the three aspects, which is defined as follows:

$$\mathcal{L}_{\text{total}}\left(\theta\right) = \mathcal{L}_1 + \lambda_1\mathcal{L}_{\text{SAM}} + \lambda_2\mathcal{L}_{\text{gra}} \tag{29}$$

$$\mathcal{L}_1\left(\theta\right) = \frac{1}{n}\sum_{u=1}^{n}\left\|Y_u - X_u\right\|_1 \tag{30}$$

$$\mathcal{L}_{\text{SAM}}\left(\theta\right) = \frac{1}{n}\sum_{u=1}^{n}\arccos\left(\frac{Y_u \cdot X_u}{\left\|Y_u\right\|_2\left\|X_u\right\|_2}\right) \tag{31}$$

$$\mathcal{L}_{\text{gra}}\left(\theta\right) = \frac{1}{n}\sum_{u=1}^{n}\left\|M\left(Y_u\right) - M\left(X_u\right)\right\|_1 \tag{32}$$

where n denotes the number of images in a training batch, X_u and Y_u represent the uth input and output HSIs, respectively. $M(\cdot)$ denotes the horizontal, vertical, and spectral gradients of the input. λ_1 and λ_2 are the balance hyperparameters.

IV. Experiments and Results

A. Datasets

In our experiments, three representative public hyperspectral datasets are exploited for performance evaluation, namely Chikusei [41], Houston [42], and Pavia Centre [43]. Moreover, considering that real-world HSIs contain noise and various uncontrollable disturbances, we conducted specific experiments on real data from the Pavia University dataset [43] to investigate the robustness of the proposed model.

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1) Chikusei Dataset: The Chikusei dataset contains images with a spatial resolution of 2517 × 2335 pixels and a spectral range from 363 to 1018 nm, comprising 128 spectral bands. These HSIs were captured using the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei City, Ibaraki Prefecture, Japan.

2) Houston Dataset: The Houston dataset includes images with a spatial resolution of 4172 × 1202 pixels and a spectral range from 380 to 1050 nm, consisting of 48 spectral bands. The data were collected by the ITRES CASI 1500 hyperspectral sensor over the University of Houston campus and its surrounding urban area in Texas, USA.

3) Pavia Centre and University Dataset: The Pavia Centre and University datasets were collected by the ROSIS sensor over Pavia, a city in northern Italy, covering a spectral range of 430 to 860 nm with 115 spectral bands. Concretely, the Pavia Centre dataset originally has a spatial resolution of 1096 × 1096 pixels. After removing water absorption bands, noisy bands, and invalid central image regions, the effective image has a spatial size of 1096 × 715 pixels and contains 102 spectral bands. For the Pavia University dataset, the original resolution is 610 × 610 pixels. Following the same preprocessing steps, the effective image size is normalized to 610 × 340 pixels and includes 103 spectral bands.

4) Dataset Preprocessing: Following the data preprocessing strategy in CST [30], we performed the cropping operator on the three datasets and divided them into the training, validation, and test sets. For the Chikusei dataset, we cropped four nonoverlapping image patches with the size of 512 × 512 × 128 from the top region of the raw image as the test set. In the remaining region, overlapping cropping was applied to augment training samples. From the cropped patches, 10% were randomly selected for validation, and the remaining 90% were used for training. When $s = 2$, patches with the size of 64 × 64 × 128 were extracted with 32 pixels overlap, and $\times s$ bicubic interpolation downsampling was applied to generate LR and corresponding HR image pairs. When $s = 4$, patches of the size 128 × 128 × 128 were acquired with 64 pixels overlap and similarly downsampling. The Houston and Pavia Centre datasets were processed in a similar manner, followed by the bilinear interpolation to generate paired training, validation, and test images. In the Houston dataset, eight image patches of size 256 × 256 × 48 were cropped from the top region of the image as the test set. The remaining area was cropped into patches based on the SR scale factor. When $s = 2$, patches with the size of $64 \times 64 \times 48$ were cropped in an overlap of 32 pixels; When $s = 4$, patches with the resolution of 128 × 128 × 48 were cropped in an overlap of 64 pixels. In the Pavia Centre dataset, four image patches of the size $256 \times 256 \times 102$ were cropped from the left side of the image for testing. The remaining area is further partitioned into patches of sizes $64 \times 64 \times 102$ and $128 \times 128 \times 102$, corresponding to s values of 2 and 4, respectively. The overlaps are set to 32 and 64 pixels for $s = 2$ and $s = 4$.

B. Experimental Details

In the WESSP-Mamba network, the number of feature channels C in the backbone is set to 128. The backbone consists of four RMM modules, and each RMM module contains 3 PGSSMB blocks ($N = 3$). In the SRM module, the block expansion coefficient expend is set to 1 ($\alpha = 1$). Considering the loss function, we set $\lambda_1 = 0.5$ and $\lambda_2 = 0.1$. During the training stage, we select the Adam optimizer with a batch size of 32 and train the network for 300 epochs. The initial learning rate is set to $1e^{-4}$ and is decayed to one-tenth of its original value after 150 epochs. To improve the robustness and generalization of the model, we enrich the training sets with random rotations and flips. All experiments are carried out on an Ubuntu 22.04.2 operating system using PyTorch 2.1.1, CUDA 11.8, and Python 3.10. The training and testing processes are deployed on an NVIDIA Tesla A100 GPU (80 G).

C. Comparison Methods and Evaluation Metrics

To comprehensively validate the performance of the proposed model, we conducted comparison experiments with several representative HSI-SR methods, including Bicubic, GDRRN [23], SSPSR [9], MSDformer [11], SRDNet [44], CST [30], MambaIRv2 [16], and MBNet [45]. Among them, Bicubic is a traditional interpolation method; GDRRN and SSPSR are based on CNN architectures; MSDformer, CST, and SRDNet are Transformer-based methods; MambaIRv2 is a Mamba-based method for natural image super-resolution, while MBNet exploits the Mamba architecture for HSI. All comparison methods were retrained and evaluated under the same dataset using the same data preprocessing and augmentation strategies to ensure fairness and comparability of the comparison results. To maintain consistent training conditions, all baselines were trained with a fixed batch size of 32 while using the full input image as the patch size. Other hyperparameters—such as convolution channels, scheduler, optimizer, training epochs, learning rate, and other related settings—strictly followed the configurations reported in the corresponding References. Overall, this work made every effort to make each baseline achieve the best performance.

For extensive performance assessment, this work adopts six popular evaluation metrics: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), SAM, correlation coefficient (CC), root mean square error (RMSE), and extended global accuracy spectrum (ERGAS). The optimal values for these metrics are $+\infty$, 1, 0, 1, 0, and 0, respectively. Specifically, PSNR and RMSE are engaged in quantifying the reconstruction accuracy and error magnitude, while SSIM assesses the preservation of structural and visual quality. SAM, CC, and ERGAS are used to evaluate the spectral consistency, correlation, and global reconstruction error between the reconstructed and original spectra.

D. Experimental Results on the Chikusei Dataset

As shown in Table I, we quantitatively compare the performance of WESSP-Mamba with other methods on the Chikusei dataset. The best results are highlighted in bold, and the second-best results are marked with underlines. For

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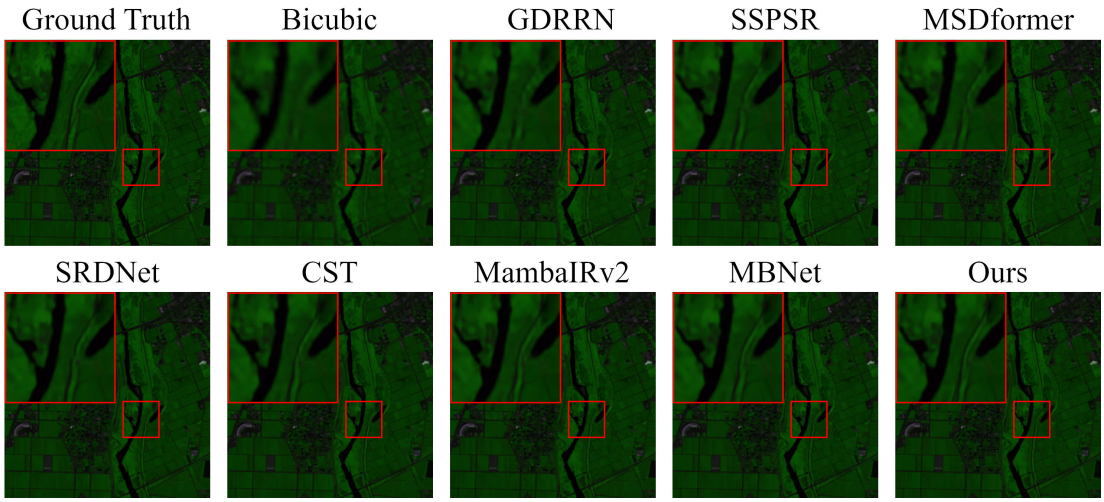


Fig. 5. Visual quality comparison on the Chikusei dataset at ×4 scale factor. Images are displayed with bands 43, 87, and 21 as R, G, and B.

TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE CHIKUSEI DATASET AT ×2 AND ×4 SCALE FACTORS. (BEST: BOLD, SECOND-BEST: UNDERLINE)

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	ERGAS↓
Bicubic	×2	43.2125	0.9721	1.7880	0.9781	0.0082	3.5981
GDRRN [24]	×2	46.3596	0.9867	1.4014	0.9883	0.0056	2.6270
SSPSR [9]	×2	47.3775	0.9892	1.2097	0.9905	0.0051	2.3236
MSDformer [11]	×2	47.6429	0.9898	1.1457	0.9910	0.0049	2.2862
SRDNet [44]	×2	-	-	-	-	-	-
CST [30]	×2	48.0203	0.9907	**1.1051**	0.9917	0.0047	2.1780
MambaIRv2 [18]	×2	47.7618	0.9901	1.2959	0.9906	0.0048	2.3835
MBNet [45]	×2	47.8120	0.9902	1.1447	0.9913	0.0048	2.2202
Ours	×2	**48.0784**	**0.9908**	1.1102	**0.9918**	**0.0047**	**2.1598**
Bicubic	×4	37.6377	0.8954	3.4040	0.9212	0.0156	6.7564
GDRRN [24]	×4	39.0765	0.9264	3.0745	0.9420	0.0130	5.8040
SSPSR [9]	×4	39.9849	0.9393	2.4880	0.9529	0.0119	5.1873
MSDformer [11]	×4	40.0388	0.9398	2.4276	0.9534	0.0118	5.1149
SRDNet [44]	×4	39.9801	0.9399	2.4426	0.9526	0.0119	5.1984
CST [30]	×4	40.1918	0.9426	2.3458	0.9549	0.0116	5.0466
MambaIRv2 [18]	×4	39.7585	0.9387	2.7279	0.9490	0.0120	5.4346
MBNet [45]	×4	40.1951	0.9423	2.3914	0.9549	0.0116	5.0487
Ours	×4	**40.2559**	**0.9437**	**2.3214**	**0.9554**	**0.0115**	**5.0241**

methods that cannot be executed in the current setting, the corresponding experimental results are represented by a dash "–" in the Table I.

The experimental results at ×2 and ×4 scales manifest that the proposed model achieves competitive performance across six evaluation metrics. At the ×4 scale, WESSP-Mamba achieves improvements of 0.0608, 0.0011, and 0.0244 in PSNR, SSIM, and SAM, respectively, over the second-best method. At the ×2 scale, it still shows salient advantages in PSNR, SSIM, and SAM, while maintaining comparable performance on the remaining metrics. It can be inferred that the proposed approach offers significant advantages in maintaining spatial–spectral consistency, preserving image structure, and ensuring spectral fidelity. It is worth noting that at the ×2 scale, the SAM value of the proposed model is slightly inferior to that of the best approach (CST). It may be attributed to wavelet-based frequency modulation, which makes spectral vectors prone to slight angular deviations at this scale, thus resulting in a lower SAM value compared to CST.

To better manifest the reconstruction effectiveness of WESSP-Mamba, we conduct visual comparisons of reconstructed images, error maps, and spectral error curves with other networks at the ×4 scale, as shown in Figs. 5–7(a), respectively. Specifically, the reconstructed images are visualized by selecting the 43rd, 87th, and 21st spectral bands of the HSI as the R (red), G (green), and B (blue) channels, respectively. The error maps are generated by averaging the per-pixel errors across all spectral channels. In the visualization, colors range from blue (indicating low error) to red (indicating high error). Spectral error curves are obtained by computing the mean error across spectral bands.

As observed from the restored images, our method contains sharper and more accurate details, particularly in regions such as roads and rivers. The reconstructed results closely resemble the textures in the ground-truth images, thereby exhibiting superior visual realism. In contrast, earlier models such as Bicubic, GDRRN, and SSPSR tend to produce more pronounced blurring artifacts. Although MSDformer, SRDNet, CST, MambaIRv2, and MBNet yield relatively clearer reconstructions in certain details, their performance on road structures remains inferior to ours. In comparison, WESSP-Mamba produces more continuous and realistic road structures, indicating its merit in texture detail restoration. Regarding the error maps, the proposed model exhibits predominantly blue regions, suggesting relatively low overall reconstruction errors. Compared with other methods, WESSP-Mamba effectively reduces the extent of high-error regions. Specifically, Bicubic and CNN-based methods often generate large areas of high errors, while Transformer- and Mamba-based endeavors alleviate this issue to some degree but still fall short of WESSP-Mamba in terms of stability and overall error suppression. These results further confirm the effectiveness of our approach in recovering details. From the spectral error curves, the reconstruction spectra of WESSP-Mamba are the

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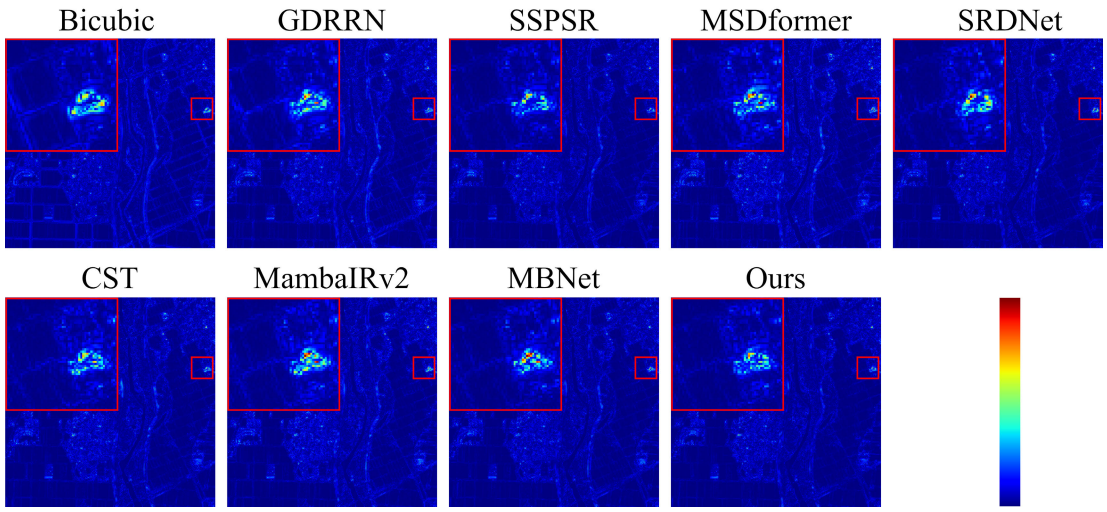


Fig. 6. Error maps of different methods on the Chikusei dataset at $\times 4$ scale factor.

TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE HOUSTON DATASET AT $\times 2$ AND $\times 4$ SCALE FACTORS. (BEST: BOLD, SECOND BEST: UNDERLINE)

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	ERGAS↓
Bicubic	×2	49.4735	0.9915	1.2707	0.9940	0.0040	1.3755
GDRRN [24]	×2	51.5081	0.9949	1.1254	0.9957	0.0031	1.0736
SSPSR [9]	×2	47.3775	0.9892	1.2097	0.9905	0.0051	2.3236
MSDformer [11]	×2	47.6429	0.9898	1.1457	0.9910	0.0049	2.2862
SRDNet [44]	×2	53.0660	0.9963	0.9297	0.9972	0.0026	0.8995
CST [30]	×2	53.2410	**0.9965**	**0.8939**	**0.9973**	0.0026	0.8818
MambaIRv2 [18]	×2	52.5320	0.9960	1.0252	0.9968	0.0028	0.9536
MBNet [45]	×2	52.7065	0.9960	0.9574	0.9969	0.0027	0.9381
Ours	×2	**53.2889**	0.9965	0.8996	0.9973	**0.0025**	**0.8749**
Bicubic	×4	43.0272	0.9613	2.5453	0.9741	0.0086	2.9085
GDRRN [24]	×4	44.1138	0.9424	2.6190	0.9752	0.0070	2.5160
SSPSR [9]	×4	45.6900	0.9783	1.9393	0.9853	0.0062	2.1158
MSDformer [11]	×4	46.2154	0.9807	1.7766	0.9870	0.0058	1.9919
SRDNet [44]	×4	45.9955	0.9797	1.8182	0.9865	0.0060	2.0462
CST [30]	×4	46.9570	0.9838	1.6492	0.9891	0.0054	1.8319
MambaIRv2 [18]	×4	46.4141	0.9820	1.8639	0.9876	0.0057	1.9395
MBNet [45]	×4	46.1456	0.9803	1.8071	0.9868	0.0059	2.0113
Ours	×4	**47.0090**	**0.9839**	**1.6352**	**0.9893**	**0.0054**	**1.8204**

closest to the ground truth in terms of spectral angle, providing smaller spectral angular errors. This not only validates the superiority of WESSP-Mamba in spectral fidelity but also highlights its capability to simultaneously preserve fine spatial details and maintain spectral consistency, underscoring its comprehensive strengths over existing approaches.

E. Experimental Results on the Houston Dataset

Table II presents the quantitative comparison of WESSP-Mamba with other approaches on the Houston dataset. At the $\times 4$ upscaling factor, our method achieves the best performance across all metrics, with performance improvements of 0.0520, 0.0001, and 0.0140 in PSNR, SSIM, and SAM, respectively, compared to the second-best method. At the $\times 2$ upscaling factor, WESSP-Mamba also exhibits strong competitiveness. Although it does not strictly surpass the second-best method in SSIM, SAM, and CC, the results are nearly equivalent. These findings indicate that WESSP-Mamba exhibits a clear merit in HSI-SR, consistently delivering high-quality reconstructions while maintaining spectral fidelity.

Following the same processing protocol as for the Chikusei dataset, we conduct a visualization analysis on the Houston dataset at a $\times 4$ scale, with the results shown in Figs. 7(b)–9. The reconstructed images are displayed using the 29th, 26th, and 19th spectral bands as the R-G-B channels. In the reconstructed results, WESSP-Mamba stands out among all relative methods, especially in complex urban scenes. It preserves stronger structural continuity along building edges and provides richer texture details in vegetation regions, whereas other networks tend to suffer from texture blurring or detail loss. The error maps further highlight these differences. Concretely, Bicubic, GDRRN, and SSPSR exhibit significant errors along building edges and large patches of artifacts in vegetation areas. CST, SRDNet, MBNet, and other methods show overall lower error levels (with error maps mostly appearing blue), but still suffer from noticeable inaccuracies in the texture details of building edges. In contrast, our model also has some errors, but they are generally smaller in magnitude, with fewer red error regions along building boundaries. From the spectral curves, the reconstruction outcomes of WESSP-Mamba consistently lie at the lowest position among all error curves, showing the smallest deviation from the ground-truth spectral curves. This further verifies its effectiveness in preserving spectral fidelity.

F. Experimental Results on the Pavia Centre Dataset

In Table III, we report a quantitative comparison of WESSP-Mamba with other methods on the Pavia Centre dataset. It is evident that our method achieves the best performance across all six metrics for both $\times 2$ and $\times 4$ upscaling factors, highlighting its superior SR capability.

Specifically, at the $\times 2$ scale, WESSP-Mamba improves PSNR, SSIM, and SAM by 0.1338, 0.0011, and 0.0528,

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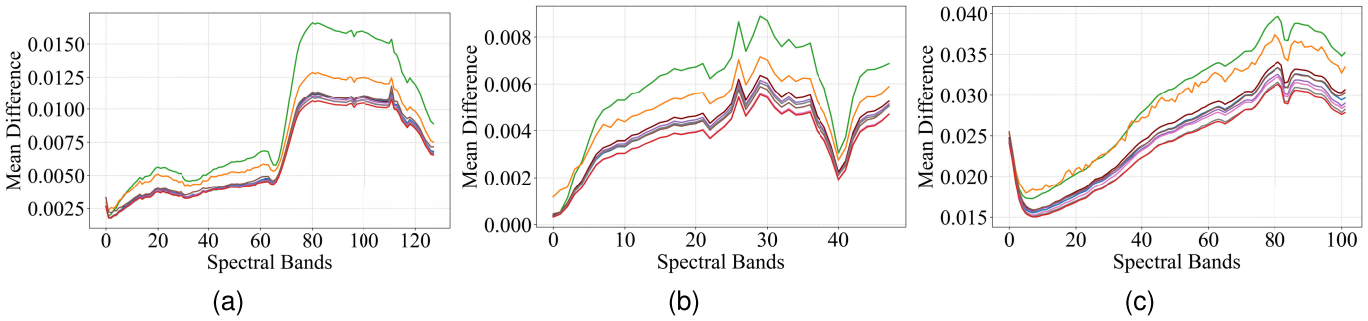


Fig. 7. Spectral error curves of different methods at a ×4 scale factor on three datasets. (a) Chikusei dataset. (b) Houston dataset. (c) Pavia Centre dataset.

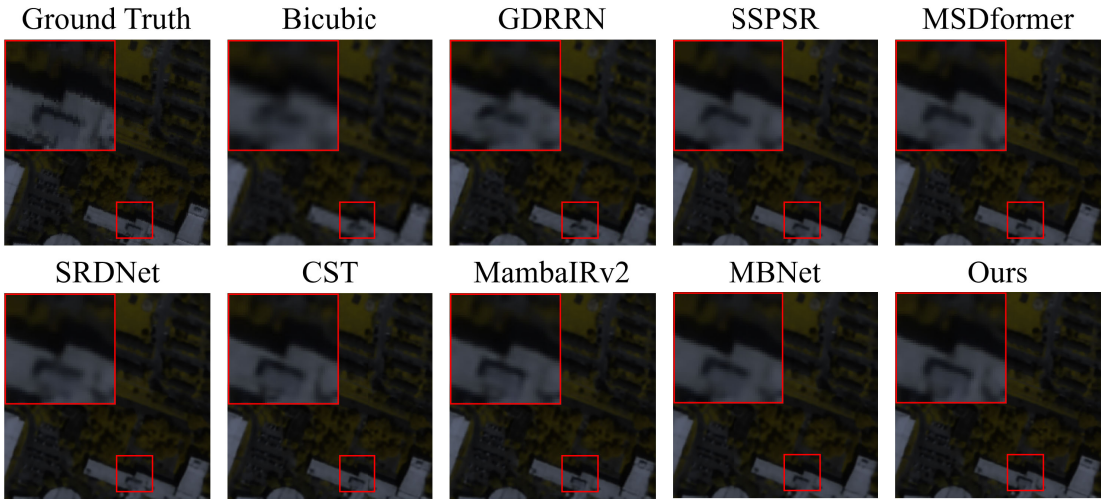


Fig. 8. Visual quality comparison on the Houston dataset at ×4 scale factor. Images are displayed with bands 29, 26, and 19 as R, G, and B.

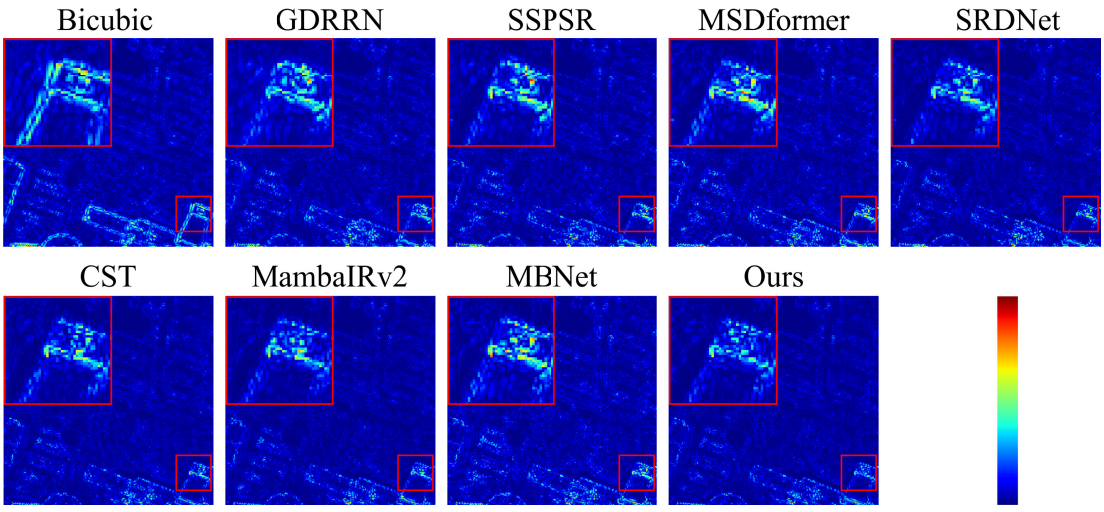


Fig. 9. Error maps of different methods on the Houston dataset at ×4 scale factor.

respectively, compared with the second-best method. At the ×4 scale, the corresponding improvements are 0.0443, 0.0042, and 0.2238. These results further confirm the reliability and effectiveness of our method in HSI-SR tasks, demonstrating its ability to consistently enhance reconstruction quality across different upscaling factors. Figs. 7(c), 10, and 11 present the reconstructed images, error maps, and spectral error curves of different methods on the Pavia Centre dataset, respectively. The recovered images are visualized using the 29th, 26th, and 19th spectral bands as the R-G-B channels.

We can observe that our approach accurately restores the image details and structural features, such as rooftop chimneys and the textures along the road edges. The error

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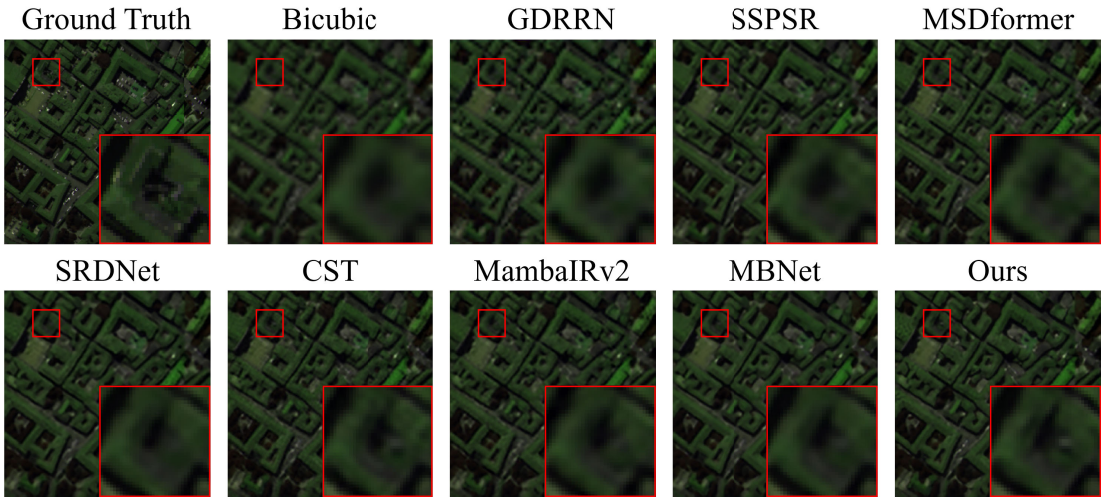


Fig. 10. Visual quality comparison on the Pavia Centre dataset at $\times 4$ scale factor. Images are displayed with bands 30, 45, and 12 as R, G, and B.

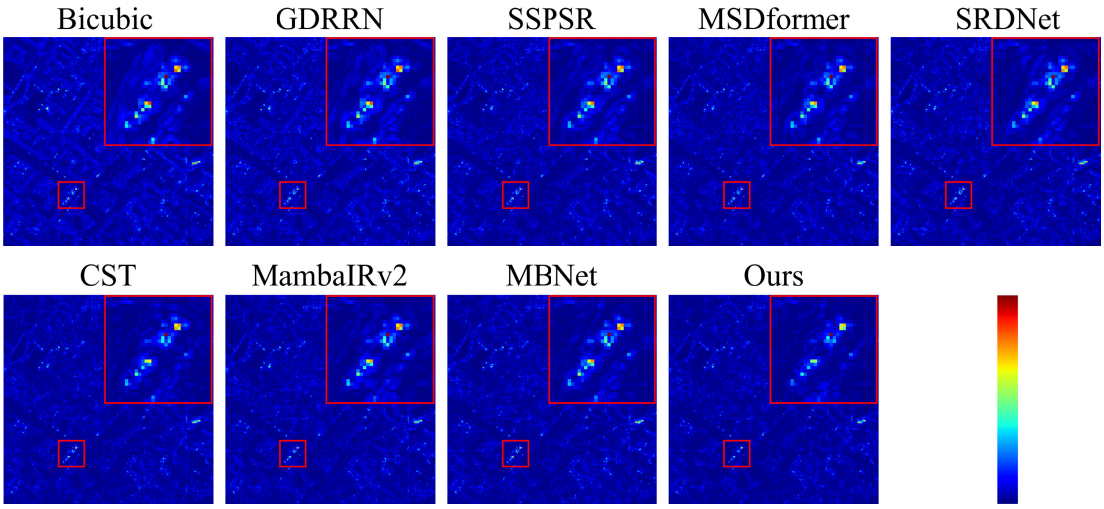


Fig. 11. Error maps of different methods on the Pavia Centre dataset at $\times 4$ scale factor.

maps further indicate that our model exhibits smaller overall reconstruction errors in comparison with other schemes, particularly in regions with complex structures or high contrast. The spectral error curves show that WESSP-Mamba maintains the lowest errors across all spectral bands, validating its superiority in preserving spectral fidelity. Overall, these results clearly prove that our method achieves a good balance between local detail recovery and global structural preservation, which can be clearly observed in the visual quality and the distribution of reconstruction errors.

In summary, the outstanding performance of WESSP-Mamba across the three datasets extensively proves its strengths in modeling both local and global spatial–spectral relationships, as well as its capability to preserve fine details. However, our training assumes a bicubic downsampling degradation, which is different from sensor blur and noise in real-world applications. Therefore, an interesting work would be to test the proposed method on real sensor data or incorporate robust degradation models.

G. Experimental Results on Real Dataset

In real scenarios, the LR HSIs obtained directly from sensors involve more complex structural characteristics than those generated by simple interpolation. To investigate the applicability of WESSP-Mamba in real-world scenarios, we reference the processing protocol described in [12], [46], and [47] and conduct real-data experiments on WESSP-Mamba and the competing methods. We first remove the last spectral band from the Pavia University dataset and crop a 256×256 patch. This patch is then fed into the $\times 2$ SR model trained on the Pavia Centre dataset. From this perspective, this is a cross-dataset testing in that all models are trained on the Pavia Centre dataset and tested on the Pavia University dataset. Subsequently, we make quantitative and qualitative analyses of the reconstructed images. For the quantitative assessment, the no-reference hyperspectral image quality metric (NRQMS) is exploited from the work in [48], where lower scores indicate better quality. In addition, we also measure the inference time

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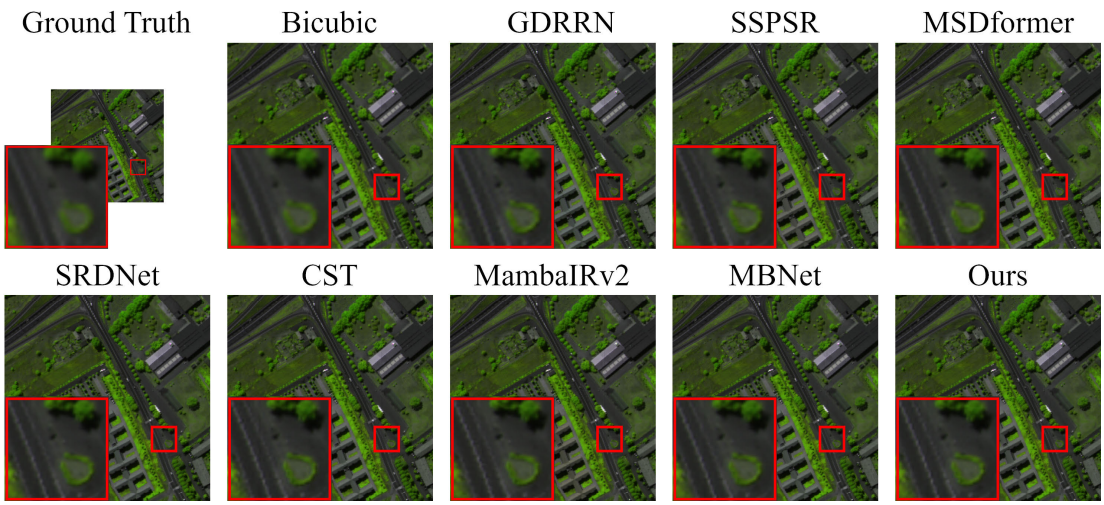


Fig. 12. Visual quality comparison on the Pavia University dataset at $\times 2$ scale factor. Images are displayed with bands 74, 82, and 65 as R, G, and B.

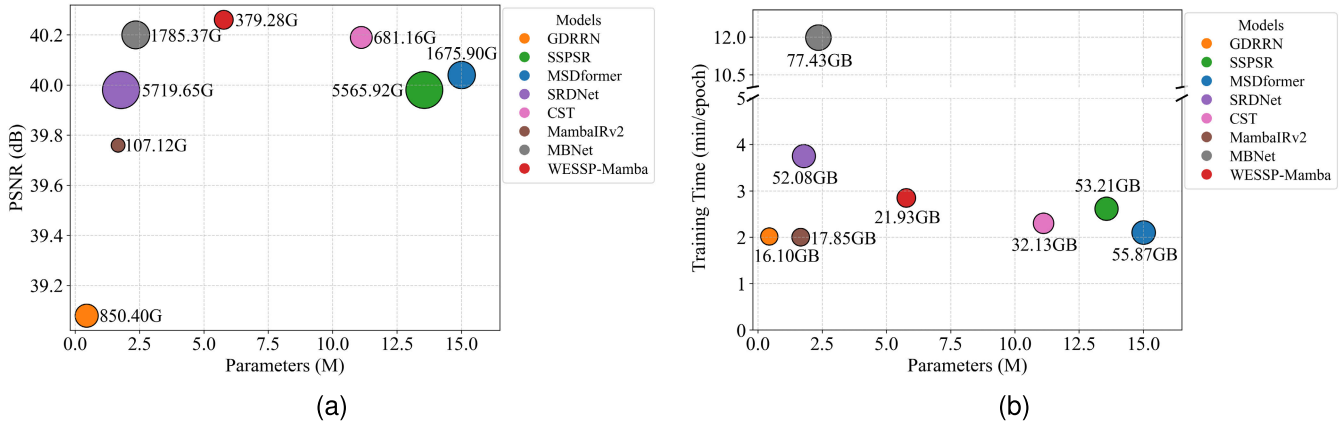


Fig. 13. Model complexity comparison of different models on the Chikusei dataset at $\times 4$ scale factor. (a) Comparison of PSNR, parameters, and FLOPs. (b) Comparison of training time, parameters, and memory usage.

of each model, in terms of the time from the input patch to the reconstructed image, with results summarized in Table IV. For qualitative comparison, we visualize the reconstructed images by mapping the 74th, 82nd, and 65th spectral bands to the RGB channels, as shown in Fig. 12. The results demonstrate that our method achieves superior quantitative performance and produces clearer reconstructed images than competing approaches, indicating strong robustness and the ability to restore higher-quality HSIs in real-world scenarios. Nonetheless, the inference time of our method is not very efficient and needs further optimization.

H. Model Complexity

To better evaluate complexity and computational efficiency, we record the model parameters, computational cost (FLOPs), GPU memory usage, and the training time per epoch for different models using an NVIDIA A100 GPU with a batch size of 32 on the Chikusei dataset for the $\times 4$ SR task. Based on this configuration, we give two joint visualizations: (a) PSNR together with FLOPs and model parameters, and (b) training time and GPU memory usage together with model parameters, as shown in Fig. 13. From Fig. 13, GDRRN, as an early method, exhibits the lowest computational overhead and resource consumption but falls short in reconstruction image quality. Schemes like SSPSR, MSDformer, SRDNet, CST, and MBNet improve performance by incorporating specific complex frameworks, at the cost of heavy resource demands. Although MambaIRv2 is lightweight in terms of computational cost compared to RGB image SR methods, its performance remains somewhat limited in hyperspectral scenarios. Notably, WESSP-Mamba achieves superior reconstruction quality while maintaining relatively low computational and memory requirements. Briefly, our model obtains a good balance between SR performance and computational efficiency.

I. Ablation Study

In this section, we conduct a series of ablation experiments based on the $\times 4$ super-resolution task on the representative Chikusei dataset to verify the effectiveness of the key modules and parameters in the proposed network framework.

1) Structural Effectiveness: Considering the overall structure of WESSP-Mamba, we deliberately investigate the impact

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TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE PAVIA CENTRE DATASET AT ×2 AND ×4 SCALE FACTORS. (BEST: BOLD, SECOND BEST: UNDERLINE)

Method	Scale	PSNR↑	SSIM↑	SAM↓	CC↑	RMSE↓	ERGAS↓
Bicubic	×2	32.1036	0.9122	4.6025	0.9549	0.0254	4.2553
GDRRN [24]	×2	33.6203	0.9376	4.9461	0.9663	0.0212	3.6251
SSPSR [9]	×2	34.8707	0.9514	4.0962	0.9741	0.0184	3.1577
MSDformer [11]	×2	35.6582	0.9596	3.6837	0.9780	0.0168	2.9045
SRDNet [44]	×2	35.7264	0.9593	3.6848	0.9782	0.0167	2.8839
CST [30]	×2	35.9798	0.9619	3.6068	0.9793	0.0162	2.8119
MambaIRv2 [18]	×2	35.6178	0.9572	3.9831	0.9766	0.0169	2.9651
MBNet [45]	×2	35.7976	0.9600	3.6750	0.9786	0.0166	2.8579
Ours	×2	**36.1136**	**0.9630**	**3.5540**	**0.9798**	**0.0160**	**2.7745**
Bicubic	×4	27.5535	0.7233	6.5161	0.8717	0.0436	7.0672
GDRRN [24]	×4	27.8101	0.7541	7.7605	0.8764	0.0420	6.8907
SSPSR [9]	×4	28.5571	0.7903	6.3245	0.8966	0.0386	6.3168
MSDformer [11]	×4	28.7162	0.7968	5.9437	0.9004	0.0379	6.2098
SRDNet [44]	×4	28.7274	0.7973	6.1454	0.9003	0.0377	6.2112
CST [30]	×4	28.9266	0.8072	5.8932	0.9053	0.0370	6.0607
MambaIRv2 [18]	×4	28.6902	0.7977	6.4571	0.8992	0.0379	6.2368
MBNet [45]	×4	28.9693	0.8103	6.0000	0.9060	0.0367	6.0365
Ours	×4	**29.0136**	**0.8145**	**5.6694**	**0.9071**	**0.0366**	**6.0011**

of the SRM module, the prior module, and the PFB module on SR performance. The quantitative results are reported in Table V. To ensure fairness, the comparative experiments are performed by removing or replacing the corresponding modules while keeping the rest of the architecture unchanged. Specifically, the structure (a) removes the Shuffle branch in the SRM module; the structure (b) removes the PFB module, i.e., no prior information; the structure (c) replaces the SSPrior module with a standard 3×3 convolution module; the structure (d) reinforces both high- and low-frequency bands of DWT in the SSPrior block using the same convolution operators; the structure (e) replaces the PFB module with a general convolution addition operation; the structure (f) replaces the PFB module with a residual spatial attention module; the structure (g) is the complete model with all modules.

From the comparison results in Table V, it can be seen that removing or replacing these modules leads to variable degrees of performance degradation, indicating that the SRM module, prior module, and PFB module consistently make critical contributions in the reconstruction process. Their collaborative design effectively improves the SR restoration quality of the complete model, confirming the rationality and effectiveness of our proposed network structure. In particular, from the result of architecture (d) in Table V, it is clear that the same enhancement mean results in a decline in the quality of the reconstruction, which demonstrates the need for variable processing mechanisms for the corresponding frequency bands. When the SSPrior is produced via traditional convolution operations or in our learnable DWT manner, it inevitably incurs minor parameters and computational cost. Notably, the gains in reconstruction performance make the additional model complexity negligible.

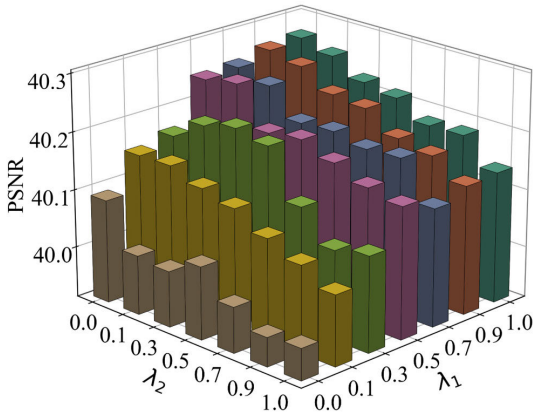


Fig. 14. PSNR values for different combinations of λ_1 and λ_2 at a ×4 scale factor on the Chikusei dataset.

2) Number of PGSSMB: The number of PGSSMB modules is one of the key factors in the performance. Massive modules significantly increase computational resource consumption and may suffer from overfitting problems, while few modules hinder the representation capability, resulting in decreased network performance. The related experimental results are reported in Table VI.

It can be observed that as the number of modules increases, the evaluation metrics of the reconstructed images gradually improve. However, when the number reaches a certain point, the performance begins to decline. Considering both performance and efficiency, we set the number of PGSSMB modules to 3 to achieve an optimal balance between reconstruction quality and computational cost.

3) Number of RMM: The number of RMM modules also has a pivotal impact on the model performance. An excessive number of RMM modules causes high computational cost, whereas too few RMM modules restrict the feature modeling capability. As reported in Table VII, the network attains the best PSNR with 4 RMM modules, whereas several other metrics achieve slightly higher values at 5 RMM modules. Given the marginal performance difference between the two settings, we set the optimal number of RMM modules to 4 to cut down computational cost while maintaining competitive performance.

4) Loss Function: To further investigate the specific contributions of each loss component during the model training stage, we conducted ablation experiments focused on the loss function. By progressively adjusting combinations of different loss functions, we analyzed the specific impacts on the SR outcome. Fig. 14 presents the ablation results on variable loss functions with the grid search strategy. The experimental results indicate that when λ_2 is fixed, and λ_1 gradually increases from 0 to 1, the reconstruction quality steadily enhances and becomes stable once λ_1 reaches 0.5. In contrast, when λ_1 is fixed while λ_2 increases, the overall reconstruction quality exhibits a declining trend. These findings suggest that integration of the SAM loss to the basic loss function significantly boosts model performance, while the introduction of an alone gradient loss incurs a slight decline. It is also observed that when all three loss components are jointly applied, a

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TABLE IV

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET AT ×2 SCALE FACTOR. (BEST: BOLD, SECOND BEST: UNDERLINE)

Method	Bicubic	GDRRN [24]	SSPSR [9]	MSDformer [11]	SRDNet [44]	CST [30]	MambaIRv2 [18]	MBNet [45]	Ours
NRQMS↓	16.0029	15.8193	15.5613	15.1874	15.1114	<u>14.8298</u>	14.8600	14.9730	**14.7634**
Inference time(s)	-	**0.3158**	<u>0.5466</u>	0.7381	0.8532	0.7450	0.8553	1.5955	1.2745

TABLE V

ABLATION STUDY ON STRUCTURAL EFFECTIVENESS AT ×4 SCALE FACTOR ON THE CHIKUSEI DATASET (BEST: BOLD)

Architectures	Params. ($\times 10^6$)	FLOPs ($\times 10^9$)	PSNR↑	SSIM↑	SAM↓
(a) SRM w/o Shuflle	5.7736	11.8462	40.2227	0.9437	2.3286
(b) PGSSMB w/o PFB	4.7332	11.1705	40.1550	0.9431	2.4113
(c) Replacing SSPrior with Conv	5.3287	11.7769	40.2178	0.9437	**2.2964**
(d) SSPrior HF/LF Conv	6.0688	11.9280	40.2269	0.9432	2.3385
(e) Replacing PFB with Conv	5.6714	11.7495	40.2014	0.9428	2.3338
(f) Replacing PFB with Residual Attention	5.4745	11.5493	40.2480	0.9437	2.3316
(g)WESSP-Mamba	5.7736	11.8525	**40.2559**	**0.9437**	2.3214

TABLE VI

ABLATION STUDY ON THE NUMBER OF PGSSMB MODULES AT ×4 SCALE FACTOR ON THE CHIKUSEI DATASET (BEST: BOLD)

Parameter value	Params. ($\times 10^6$)	FLOPs ($\times 10^9$)	PSNR↑	SSIM↑	SAM↓
N=1	4.0906	10.1417	40.1538	0.9424	2.3496
N=2	4.9321	10.9971	40.1805	0.9431	2.3386
N=3	5.7736	11.8525	**40.2559**	**0.9437**	**2.3214**
N=4	6.6152	12.7079	40.1598	0.9430	2.3404

TABLE VII

ABLATION STUDY ON THE NUMBER OF RMM MODULES AT ×4 SCALE FACTOR ON THE CHIKUSEI DATASET (BEST: BOLD)

Parameter value	Params. ($\times 10^6$)	FLOPs ($\times 10^9$)	PSNR↑	SSIM↑	SAM↓
RMM=2	4.2162	10.2674	40.2280	0.9435	2.3253
RMM=3	4.9949	11.0600	40.2417	0.9437	2.3230
RMM=4	5.7736	11.8525	**40.2559**	0.9437	2.3214
RMM=5	6.5524	12.6451	40.2523	**0.9437**	**2.3196**

minor performance gain occurs at $\lambda_2 = 0.1$. Accordingly, the λ_1, corresponding to the highest PSNR, is chosen under the condition of $\lambda_2 = 0.1$, i.e., $\lambda_1 = 0.5$ and $\lambda_2 = 0.1$. Taken together, the model achieves salient performance gains when all three loss components function simultaneously, confirming the complementary strengths of the three loss components for HSI-SR tasks.

5) Residual Modulation Coefficient: To inspect the contributions of three residual modulation coefficients, we conduct ablation experiments by enabling them individually and in pairwise combinations. The results are listed in Table VIII.

We can find that the network performance degrades profoundly without any residual modulation coefficients. When only a single coefficient is enabled, the performance still decreases. When adopting pairwise combinations, the network performance remains inferior to the full configuration. To sum up, the network achieves the best performance only when all three coefficients are activated simultaneously, which confirms the optimal configuration of residual modulation coefficients.

TABLE VIII

ABLATION STUDY ON RESIDUAL MODULATION COEFFICIENT AT ×4 SCALE FACTOR ON THE CHIKUSEI DATASET (BEST: BOLD)

α_1	α_2	α_3	PSNR↑	SSIM↑	SAM↓
×	×	×	39.8743	0.9387	2.4215
✓	×	×	39.9683	0.9397	2.4070
×	✓	×	40.1805	0.9428	2.3340
×	×	✓	39.9350	0.9397	2.4300
✓	✓	×	40.2085	0.9429	2.3322
✓	×	✓	40.0994	0.9421	2.3831
×	✓	✓	40.2495	0.9437	2.3221
✓	✓	✓	**40.2559**	**0.9437**	**2.3214**

V. CONCLUSION

This article proposes an efficient WESSP-Mamba model for HSI-SR. We devise the SRM module by incorporating a shuffle branch into the Mamba network to improve spatial modeling and exploit a spectral attention mechanism to fully prioritize spectral relations. To mitigate detail loss during feature propagation, the model leverages DWT and spectral attention to form informative priors. Furthermore, the PFB is embedded to adaptively fuse prior information with deep features in a cross-layer manner. Extensive experimental results demonstrate that WESSP-Mamba outperforms mainstream HSI-SR methods in reconstruction accuracy, structural detail preservation, and spectral consistency, achieving better overall performance of SR. However, WESSP-Mamba critically relies on the accuracy and stability of prior information. The DWT used for prior information involves a fixed wavelet basis function, which cannot adaptively respond to the input HSI. This fixed basis struggles to represent discriminative features across diverse data distributions, hence degrading the prior quality and the final reconstruction performance.

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In future work, we will explore more efficient prior extraction techniques and investigate the integration of diffusion models to enhance reconstruction quality. Additionally, the adaptability of the Mamba module for 3-D HSI modeling from a structural perspective will be probed to better leverage its potential for long-range spatial–spectral dependency learning.

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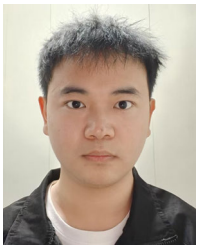
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